Exploiting The Dual-Tree Complex Wavelet Transform for Ship Wake Detection in SAR Imagery

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Abstract

1. Detecting ships and their corresponding wakes has always been a significant topic in terms of marine applications.
2. In this study, the ship wake detection problem is addressed as a Radon transform based inverse problem with **Cauchy and DT-CWT** regularizations.
3. The results show a successful detection performance of up to **90.91**%

The Proposed Method

Since ship wakes have linear features, SAR image formation can be expressed in terms of its Radon transform as

$$Y = CX + N$$

Where Y is the noisy SAR image, X represents the data in the Radon domain. Also, N is additive noise, and the operator $C = R^{-1}$ is the inverse Radon transform. The problem is solved to obtain an estimation for X via the minimization problem of

$$\hat{X} = \arg\min_{X}\{\|Y - CX\|_2^2 - \log(\gamma/(\gamma^2 + X^2)) - \lambda\|\mathcal{B}X\|_1\}$$

Where $\mathcal{B}$ is forward DT-CWT operator.

Algorithm 1 is based on forward-backward (FB) splitting algorithm to obtain the optimized $\hat{X}$.

Algorithm 1: *Algorithmic representation of Cauchy and DT-CWT Proximal Splitting*

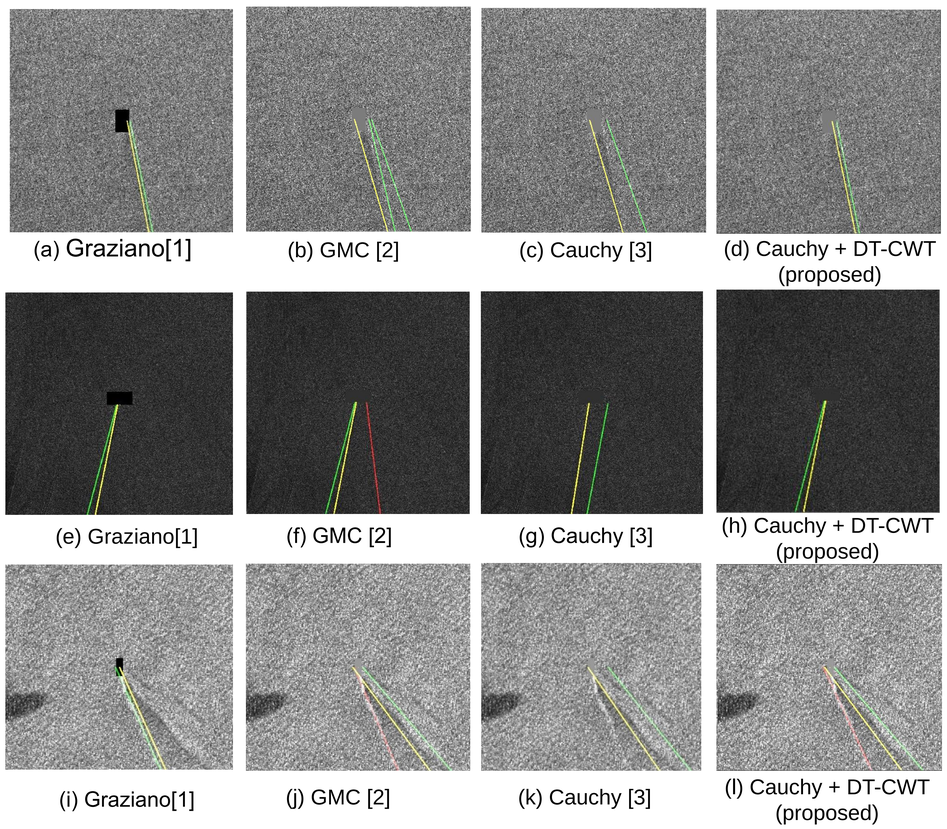
1: Input: SAR imagery Y and coefficients μ, γ, λ
2: Output: Radon image X
3: Set: $i = 0$ and $X^{(0)} = \{0\}$
4: do
5: $Z^{(i)} = X^{(i)} - \mu\left(C^T\left(C\left(X^{(i)}\right) - Y\right) - \nabla\left(log\left(\frac{\gamma}{\gamma^2 + X^{(i)2}}\right)\right)\right)$
6: $W = soft\{\mathcal{B}(Z^{(i)}),\ \mu * \lambda\}$
7: $X^{(i)} = \mathcal{B}^{-1}(W)$
8: $i++$
9: **while** $\varepsilon^{(i)} > 10^{-3}$ *or* $i < maxIter$

Why DT-CWT?

DT-CWT can significantly reduce the translation sensitivity and improve the direction selectivity of Discrete Wavelet Transform. In this case, implementing DT-CWT in Radon domain of SAR imagery would be useful to enhance linear features of ship wakes.

Detection Performance for All SAR Images

	TP	TN	FP	FN	Accuracy
Graziano et. al. [1]	32.73%	38.18%	25.45%	3.64%	70.91%
GMC [2]	**43.64%**	34.55%	18.18%	3.64%	77.27%
TV	36.18%	33.64%	26.36%	**1.82%**	70.91%
Cauchy [3]	37.27%	**47.27%**	10.00%	6.36%	84.55%
Proposed Cauchy + DTCWT	41.82%	**48.18%**	**5.45%**	5.45%	**90.91%**



(a) Graziano[1] (b) GMC [2] (c) Cauchy [3] (d) Cauchy + DT-CWT (proposed)

(e) Graziano[1] (f) GMC [2] (g) Cauchy [3] (h) Cauchy + DT-CWT (proposed)

(i) Graziano[1] (j) GMC [2] (k) Cauchy [3] (l) Cauchy + DT-CWT (proposed)

Detection Processing

Step 1:	Step 2:
Pre-processing of SAR images: Obtain centered and masked image **Y** in the spatial domain	Use inverse problem Y = CX+N to reconstruct the Radon image $\hat{X}_{\text{Cauchy-DTCWT}}$ by the Algorithm 1.
Step 3:	**Step 4:**
3.1 Restriction of reconstructed Radon images to obtain restricted images $\hat{X}_{\text{res}}$	4.1 Inverse Radon transform of detected bright(dark) points
3.2 Detect Turbulent and first Narrow-V wake in the Radon domain	4.2 Retain half lines and estimate right indexes
3.3 Detect second Narrow-V wake and Kelvin arms in the Radon domain	4.3 Validate and discard false wake detection

Reference

[1] M. Graziano, M. D'Errico and G. Rufino, "*Wake Component Detection in X-Band SAR Images for Ship Heading and Velocity Estimation*", Remote Sensing, vol. 8, no. 6, p. 498, 2016.

[2] O. Karakus, I. Rizaev, and A. Achim, "*Ship Wake Detection in SAR Images via Sparse Regularization*," IEEE Transactions on Geoscience and Remote Sensing, vol. 58, no. 3, pp. 1665–1677, 2020.

[3] O. Karakus, and A. Achim, "*On solving SAR imaging inverse problems using non-convex regularization with a Cauchy-based penalty*, "IEEE Transactions on Geoscience and Remote Sensing, 2020.